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CPCS-214 Computer Architecture and Organization

32 - Register File

Introduction

- Register file is one structure that is central to datapath

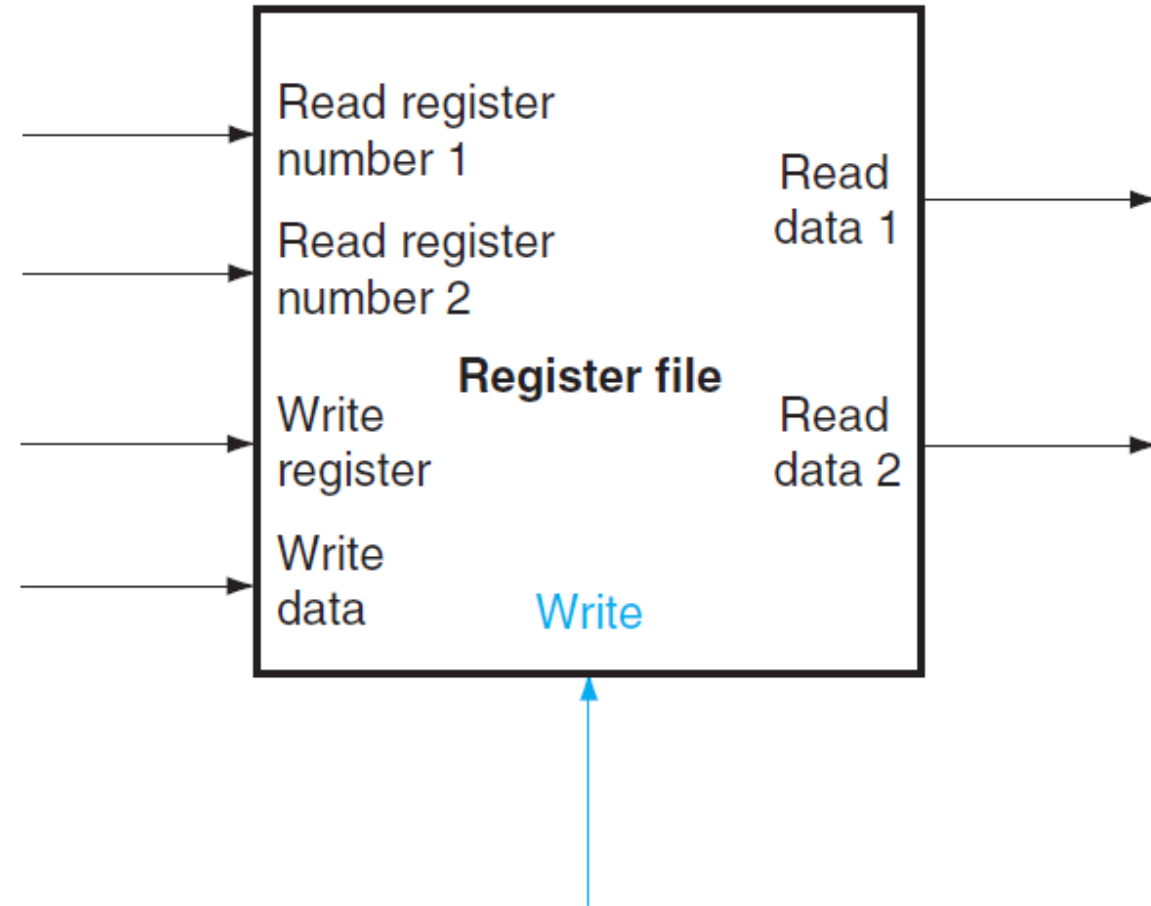
Register File

- Consists of a set of registers that can be read and written by supplying a register number to be accessed
- Can be implemented with a decoder for each read or write port and an array of registers built from **D flip-flops**

Read and Write a Register

- Because reading a register does not change any state, we need only supply a register number as an input, and the only output will be the data contained in that register
- For writing a register we will need three inputs
 - register number
 - data to write
 - clock that controls writing into register

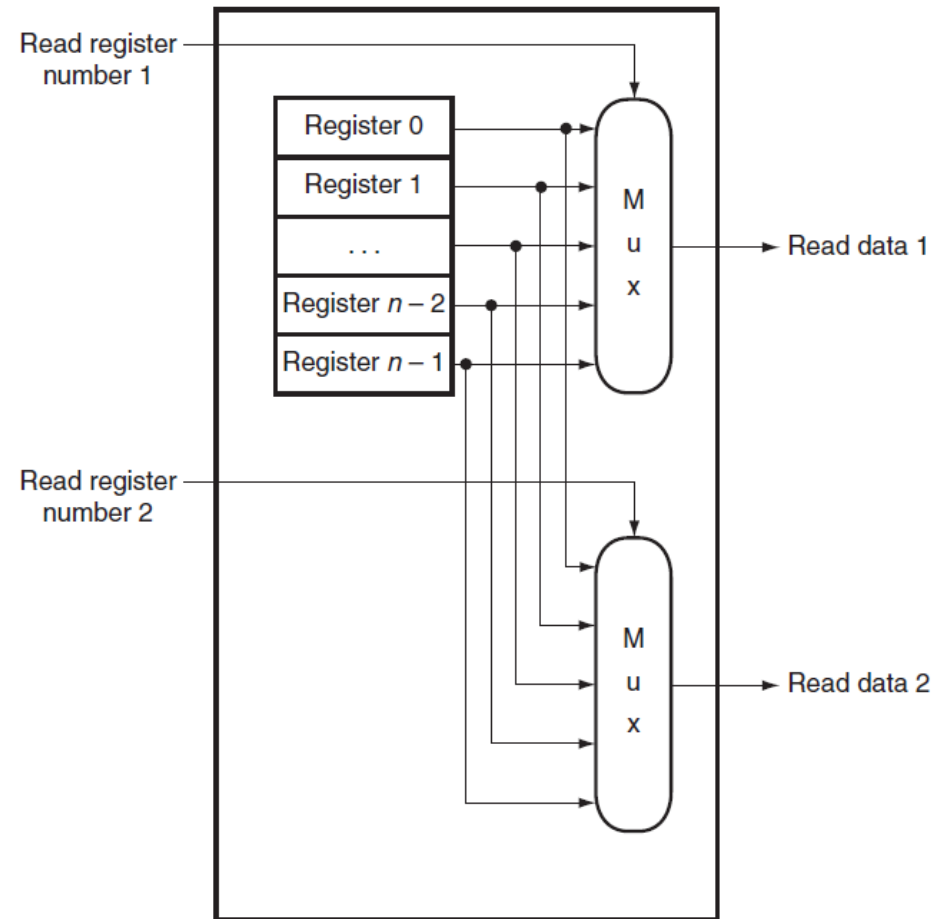
Register File



Read Ports

- Can be implemented with pair of multiplexors, each of which is as wide as number of bits in each register of the register file
- Figure shows implementation of two register read ports for a 32-bit-wide register file

Read Ports



Write Port

- Slightly more complex, since we can only change contents of designated register
- We can do this by using a decoder to generate a signal that can be used to determine which register to write
- Figure shows implementation of write port for a register file
- Important to remember that flip-flop changes state only on the clock edge

Write Port

